

Hyunseok Lee

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🎓 Google Scholar 🌐 GitHub

RESEARCH INTERESTS My research interests lie in building intelligence that is self-aware and safe. To this end, I focus on developing Large Language Models (LLMs) that can reason, make decisions (i.e., exhibit agentic behaviors), and ensure safety. I am also broadly interested in the continual pretraining of LLMs and their multilingual capabilities.

Keywords: LLM, VLM, Reasoning, LLM-based Agents, LLM Safety

EDUCATION **Ph.D. in Artificial Intelligence** Mar. 2024 - Present
Korea Advanced Institute of Science and Technology (KAIST)
Advisor: **Jinwoo Shin**

B.S. in Electrical Engineering and Computer Science (double) Mar. 2018 - Feb. 2024
Korea Advanced Institute of Science and Technology (KAIST)

WORK EXPERIENCE **Microsoft Research Vancouver**, Research Intern Apr. 2026 - Jul. 2026
with **Soheil Abbasloo** Vancouver, CAD

- Topic: LLM agent, LLM reasoning, Long horizon

Microsoft Research Asia, Research Intern Jul. 2025 - Dec. 2025
with **Soheil Abbasloo** Beijing, CN

- Topic: LLM reasoning, Reinforcement Learning

NAVER Cloud, Research Intern Feb. 2025 - Jun. 2025
with **Kang Min Yoo** Seongnam, KR

- Topic: LLM reasoning, LLM Agents, Visual LM (VLM)

PUBLICATIONS * denotes equal contribution

Preprints (available upon request)

[P3] Benchmarking Visual State Tracking in Multimodal Video Understanding
Sihyun Yu^{†*}, Nanye Ma^{†*}, Pinzhi Huang^{†*}, **Hyunseok Lee**^{*}, Shusheng Yang, June Suk Choi, Ellis Brown, Oscar Michel, Boyang Zheng, Jinwoo Shin, Saining Xie
Technical Report

[P2] Beyond Correctness: Learning Robust Reasoning via Transfer
Hyunseok Lee, Soheil Abbasloo, Jihoon Tack, Jinwoo Shin

[P1] ReGUIDE: Data Efficient GUI Grounding via Spatial Reasoning and Search
Hyunseok Lee, Jeonghoon Kim, Beomjun Kim, Jihoon Tack, Chansong Jo, Jaehong Lee, Cheonbok Park, Sookyo In, Jinwoo Shin, Kang Min Yoo

Conferences

[C3] Vision-aligned Latent Reasoning for Multi-modal Large Language Model
Byungwoo Jeon, Yoonwoo Jeong, **Hyunseok Lee**, Minsu Cho, Jinwoo Shin
ICML 2025

[C2] ReVISE: Learning to Refine at Test-Time via Intrinsic Self-Verification
Hyunseok Lee*, Seunghyuk Oh*, Jaehyung Kim, Jinwoo Shin, Jihoon Tack
ICML 2025

[C1] ReMoDetect: Reward Models Recognize Aligned LLM's Generations
Hyunseok Lee*, Jihoon Tack*, Jinwoo Shin
NeurIPS 2024
Qualcomm Innovation Fellowship

HONORS	Qualcomm Innovation Fellowship Korea 2024	
INVITED TALKS	“Large Scale LLM Training and Cloud Computing Usage” SKT Enterprise AIX CON Online (remote)	Dec. 2024
	“ReMoDetect: Reward Models Recognize Aligned LLM's Generations” Max Planck Institute for Security and Privacy (remote)	Nov. 2024
INDUSTRIAL PROJECT	Korean Multilingual LLM Training for Thesis Searching Service • LLM project with Nable Communications, the web service development company. The system will be deployed at the company's thesis searching service. • Developed a multilingual Korean LLM continually trained from Llama-3.1-8B (Korean LLM for Thesis) • Applied core LLM techniques in the system: (i) multilingual continual pretraining by entangling first language, (ii) data synthesize for thesis data to pretrain and post-train, and (iii) RAG-specific training.	Mar. 2024 - Dec. 2024
ACADEMIC ACTIVITIES	Conference Revise: NeurIPS, ICML, ICLR, ECCV Teaching Assistant , “CS101: Introduction to Programming”, KAIST	Spring & Fall 2023
TECH. SKILLS	Programming: Python, C Machine Learning: PyTorch, TensorFlow, huggingface transformers, deepspeed	
SOFTWARE	Open Source: PyTorch implementation and model • Korean LLM for Thesis Search • https://github.com/hyunseoklee-ai/ReMoDetect [C1]	
REFERENCE	Jinwoo Shin , Professor at KAIST Contact: jinwoos@kaist.ac.kr Kangmin Yoo , Research Lead at Naver Cloud Contact: kangmin.yoo@navercorp.com Jihoon Tack , Senior Researcher at Microsoft Research Contact: jihoontack@microsoft.com Soheil Abbasloo , Senior Researcher at Microsoft Research Contact: soheil.abbasloo@microsoft.com	

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